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ROLL NO 62

D20A

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- [Blockchain.com](https://blockchain.com)
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?

Satoshi Nakamoto

2. Bitcoin transactions are recorded on which data structure?

Blockchain

3. Bitcoin uses which consensus mechanism?

Proof of Work (PoW)

4. The smallest unit of Bitcoin is called?

Satoshi

5. Bitcoin is a **peer-to-peer** type of what system?

Payment

Wallets

6. A wallet stores the user's Private key.
7. The recovery phrase used in wallets is called a Seed phrase.
8. Hardware wallets store keys Offline (online/offline).
9. Bitcoin addresses are derived from Public keys.
10. What is required to sign a transaction? (public/private)

Private

Security

11. Bitcoin prevents double-spending using Blockchain .
12. Bitcoin is secured using Asymmetric cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin Bitcoin Core .
14. The process of verifying transactions is called Validation .
15. The Bitcoin network is described as Decentralized (centralized/decentralized).

Mining

16. Miners solve Cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoins .
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2,016 blocks.

20. Miners compete to find a valid block Hash (hash/signature).

Blockchain Structure

21. Each block contains a Merkle root.

22. Blocks are linked using the previous block's Hash .

23. A Bitcoin transaction contains inputs and Outputs .

24. Unspent transaction outputs are called Unspent Transaction Outputs (UTXOs) .

25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain History .

27. Bitcoin uses which protocol for communication? (TCP/P2P)

Transmission Control Protocol (TCP)

28. A light wallet uses Simplified Payment Verification (SPV) mode (SPV/full).

29. Nodes relay information across a Peer-to-Peer network.

30. The maximum Bitcoin supply is **21** (million) million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: 961,125
- Block hash:0000000000000000000010ce6c8f55b484a79efa3bcf4853efbbe459c540a98fa
- Timestamp: **August 04, 2026 11:50:06**
- Number of transactions: **4607**
- Miner/pool name: **Unknown**

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.

6f3c5d987445aa94e88c21762db8b6d61def726b697da699f2a078ca03d4e2ab

11. Number of inputs:1

12. Number of outputs:2

13. Transaction fee:

0.00004540 BTC

14. How many confirmations does it have?

2

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

1AZ7aYCT6JPLF1EZ7PgymfFtH4xK44CcCP

16. Record: 1AZ7aYCT6JPLF1EZ7PgymfFtH4xK44CcCP

- Total BTC received: 146777.84527820 BTC
- Total BTC sent: 146595.44278630 BTC
- Current balance:182.40249190 BTC
- What is the role of a full node?

A **full node** downloads and stores the entire blockchain, independently verifies all transactions and blocks according to Bitcoin's consensus rules, and relays valid transactions and blocks across the network.

List any two developer resources available on bitcoin.org.

1. **Developer Guide**
2. **Developer Examples**

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: \$64,109.14
- 24-hour % change: -2.71%
- Market capitalization:\$1.27 Trillion
- Circulating supply:20,062,306 BTC
- Total/maximum supply:21,000,000 BTC
- **Task C2: Historical Data**
 - What was the price of Bitcoin one year ago?
Approximately \$≈ \$115,070 USD
 - Note the highest and lowest prices in the last 30 days.
\$66.8K USD

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price changes rapidly due to factors such as market demand and supply, investor sentiment, regulatory announcements, macroeconomic events, and speculative trading.

List two investment risks mentioned.

High price volatility

Regulatory uncertainty

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave me the clearest understanding because it explains Bitcoin concepts in simple and easy-to-understand language. It provides well-organized information about wallets, mining, security, and blockchain, making it suitable for beginners.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions in a distributed ledger that is shared among thousands of computers (nodes). Every transaction is verified through the network's consensus mechanism before being added to a block. Each block is cryptographically linked to the previous one, making the data secure and difficult to alter.

